Virtual Machine Lab Setup

This repository contains the documentation of how I designed and configured my virtual machine environment to start my journey into hacking and cyber security.

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1. Hypervisor Selection: Oracle VM VirtualBox

For my virtualization software, I chose Oracle VM VirtualBox
Why VirtualBox?It is highly intuitive, user-friendly, and very straightforward to set up. It provides an excellent, clear visual feedback of hardware allocations, making it easy to understand my computer's hardware limitations regarding CPU cores, RAM usage, and SSD space.

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2. Operating System: Parrot OS

I decided to go with Parrot OS as my primary hacking environment.
Why Parrot OS?It comes pre-packaged with a vast array of security and penetration testing tools out of the box.
Resource Efficiency:Since my host machine is limited to 8 GB of RAM, resource management was critical. Parrot OS is remarkably lightweight compared to other distros, allowing it to run incredibly smoothly within a constrained virtual space without freezing my main system.

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3. Hardware Allocation & System Tweak

To ensure optimal performance while keeping my host machine stable, I applied the following specifications to the virtual machine:

Base RAM: 5000 MB (Allocated generously to allow smooth multitasking with heavy security tools).
Swap File Tweak: I manually configured a 2048 Mb (2 GB) swapfile directly inside Parrot OS. This acts as an emergency buffer for memory management, preventing crashes if RAM usage spikes.
Processor (vCPUs):6 vCPUs (Ensures fast processing for multi-threaded tools, password cracking, and compiling).
Storage: 25 GB (Dynamic allocation on SSD for fast read/write speeds).

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4. Network Configuration

Network Mode:NAT (Network Address Translation)
Purpose: I chose NAT to provide my virtual machine with immediate, out-of-the-box internet access for downloading tools and updates, while keeping the VM safely isolated behind VirtualBox's private engine. It prevents any accidental exposure or traffic interference with my physical local network (LAN).